

IT HOME LAB

Phase 6 & Phase 8

Azure Entra ID | VM Snapshots | Backup Solutions

Document Overview

This document covers Phase 6 (Microsoft Azure Entra ID Administration) and Phase 8 (VM Snapshot Management and Enterprise Backup Solutions) of the IT Home Lab project.

Phase 6 demonstrates cloud identity management skills using Microsoft Azure Entra ID with a Global Administrator account — practising user management, group administration, role assignment, sign-in log analysis, and cloud password resets.

Phase 8 covers VM snapshot management using VirtualBox, Windows Server Backup configuration on DC01, and a comprehensive walkthrough of Altaro Backup — the enterprise VM backup solution referenced in the target job description.

Phase 6 — Azure Entra ID	Phase 8 — Snapshots	Phase 8 — Backups
User Management	VirtualBox Snapshots	Windows Server Backup
Group Administration	Checkpoint Concepts	Altaro Backup Setup
Role Assignment (RBAC)	Snapshot Restore	Altaro Configuration
Sign-in Log Analysis	Snapshot Best Practices	Backup Scheduling
Cloud Password Reset	Snapshot vs Backup	Restore Procedures

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GitHub Portfolio | IT Support Specialist | May 2026

**PHASE
6****Microsoft Azure Entra ID Administration**

1.1 Overview

Microsoft Azure Entra ID (formerly Azure Active Directory) was accessed and administered using a UTS student account with Global Administrator privileges on the tenant Student829.onmicrosoft.com. This phase replicates real-world cloud identity management tasks performed by IT support staff in hybrid and cloud-only enterprise environments.

The skills practised in this phase directly correspond to the job requirement: "Create, manage, and support user accounts, passwords, and permissions in Active Directory, Azure AD, and Office 365."

1.2 Azure for Students — Account & Credits

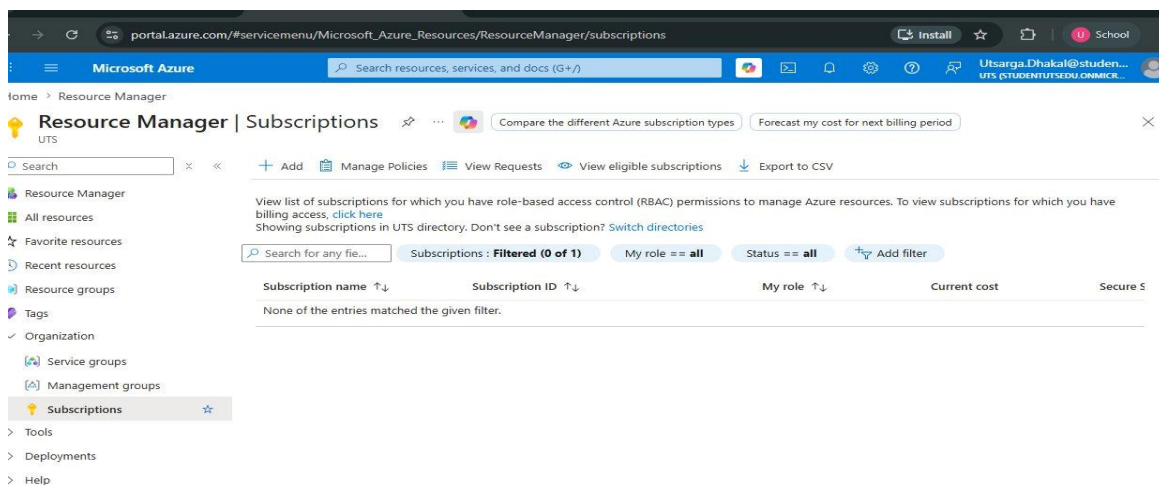


Figure 1.1: Azure for Students dashboard — \$100 USD credit active, 365 days remaining (expires 05/12/2027), \$0.00 current costs

Account Type	Tenant Name	Admin Role	License Tier
Azure for Students	Student829.onmicrosoft.com	Global Administrator	Microsoft Entra ID Free

1.3 User Management

The Microsoft 365 user management interface was used to review and manage cloud user accounts. Three users exist in the tenant, all signing in with Student829.onmicrosoft.com email addresses.

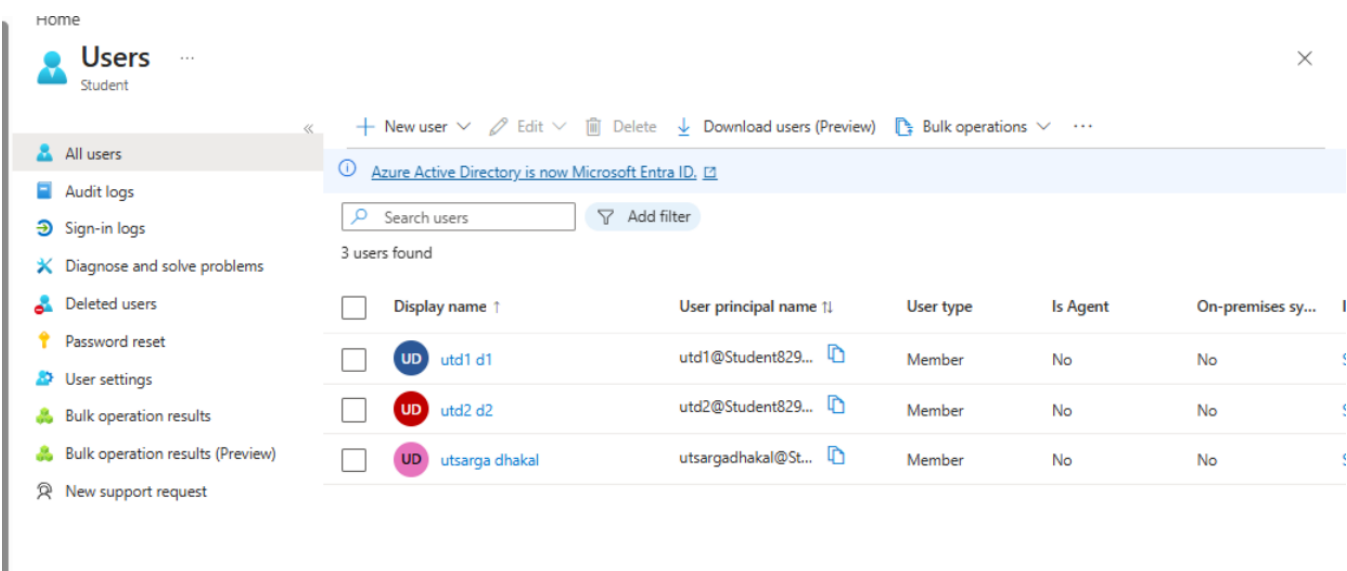


Figure 1.2: Microsoft 365 Add Users screen — existing accounts visible: utsarga dhakal (admin), utd1 d1, utd2 d2

User Account Details

Display Name	User Principal Name	User Type	Role
utsarga dhakal	utsargadhakal@Student829.onmicrosoft.com	Member	Global Admin
utd1 d1	utd1@Student829.onmicrosoft.com	Member	Standard
utd2 d2	utd2@Student829.onmicrosoft.com	Member	Standard

1.4 Group Administration

Four groups were created and managed within the Entra ID tenant. Group types were selected based on their intended purpose — Microsoft 365 groups for collaboration and Security groups for access control.

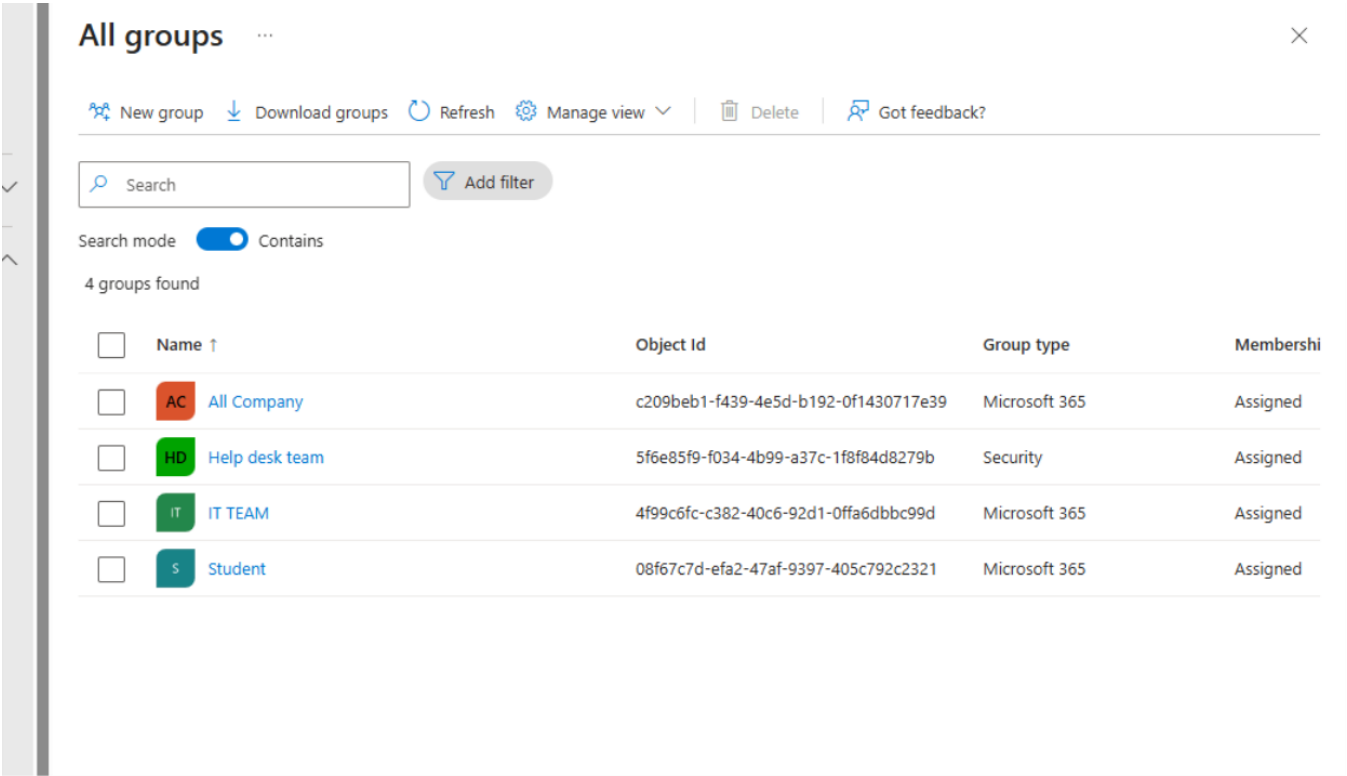


Figure 1.3: All Groups — 4 groups showing Name, Object ID, and Group Type

Group Name	Type	Purpose	Members Added
All Company	Microsoft 365	Organisation-wide communication and collaboration	All tenant users
Help desk team	Security	Access control for helpdesk resources and role assignments	utd1 d1, utsarga dhakal
IT TEAM	Microsoft 365	IT department — Teams, SharePoint, email collaboration	IT staff
Student	Microsoft 365	Student cohort group for tenant organisation	Student accounts

Group Types Explained: Microsoft 365 groups enable collaboration with shared mailbox, Teams channel, SharePoint site, and Planner. Security groups control access to resources

and applications — they are the cloud equivalent of Active Directory Security Groups used in the on-premises lab.

1.5 Role Assignment — Helpdesk Administrator (RBAC)

The Helpdesk Administrator role was assigned to utsarga dhakal, granting the ability to reset passwords for non-administrator users and manage basic user attributes across the tenant. This demonstrates Role-Based Access Control (RBAC) in Azure Entra ID.

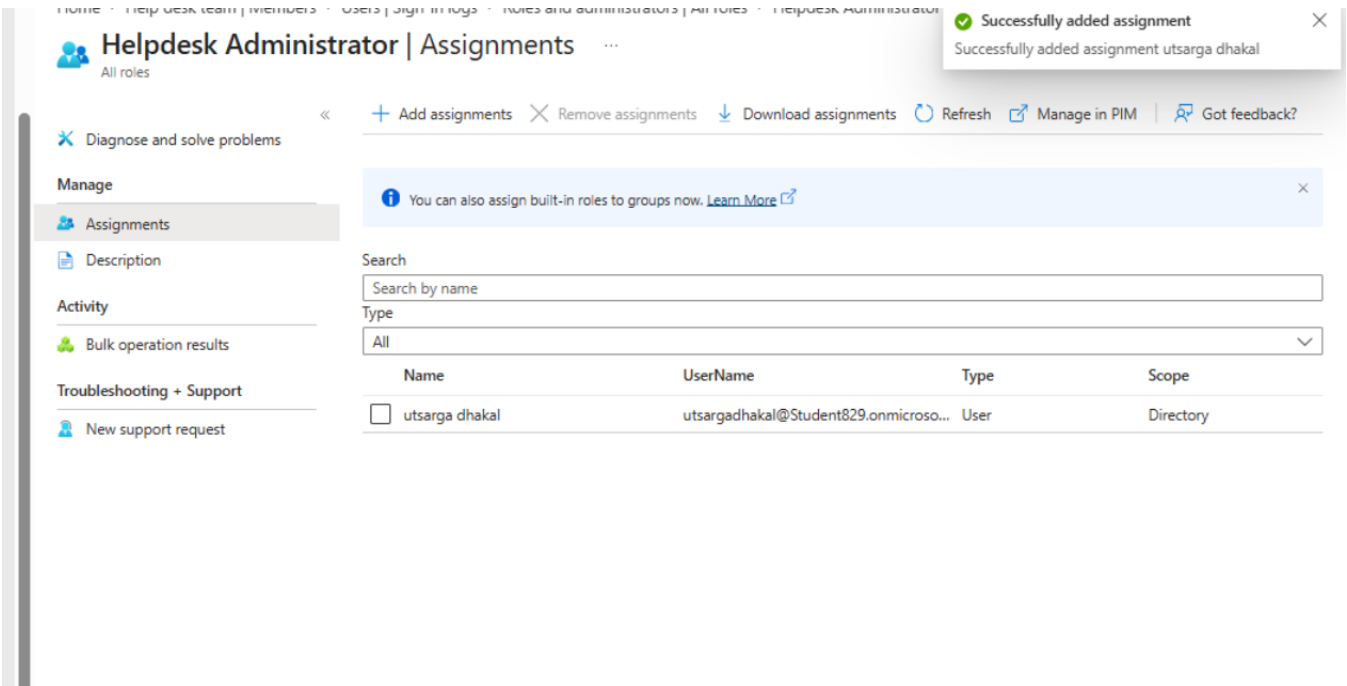


Figure 1.4: Helpdesk Administrator Assignments — utsarga dhakal successfully assigned the role

Entra ID Free (Current)	Entra ID P1/P2 (Enterprise)
Role assigned to individual users only	Role assigned to Security Groups
Manual per-user assignment required	Add user to group = instant role inheritance
Inefficient at scale (10 staff = 10 assignments)	Scalable (10 staff = 1 group assignment)
No Conditional Access Policies	Full Conditional Access (MFA by location/device)
No Self-Service Password Reset groups	SSPR enabled per group

Real-World Note: Most enterprise companies use Entra ID P1 or P2 to enable group-based role assignment (RBAC). This means when a new helpdesk analyst joins, IT simply adds them to the "Help Desk Team" group — they automatically inherit the Helpdesk Administrator role, shared mailbox access, SharePoint permissions, and any assigned applications. One action replaces many.

1.6 Sign-in Log Analysis

Security sign-in logs were reviewed in the Entra ID admin centre showing all interactive authentication events for the past 7 days. Sign-in log monitoring is a critical daily security task in IT support and security operations roles.

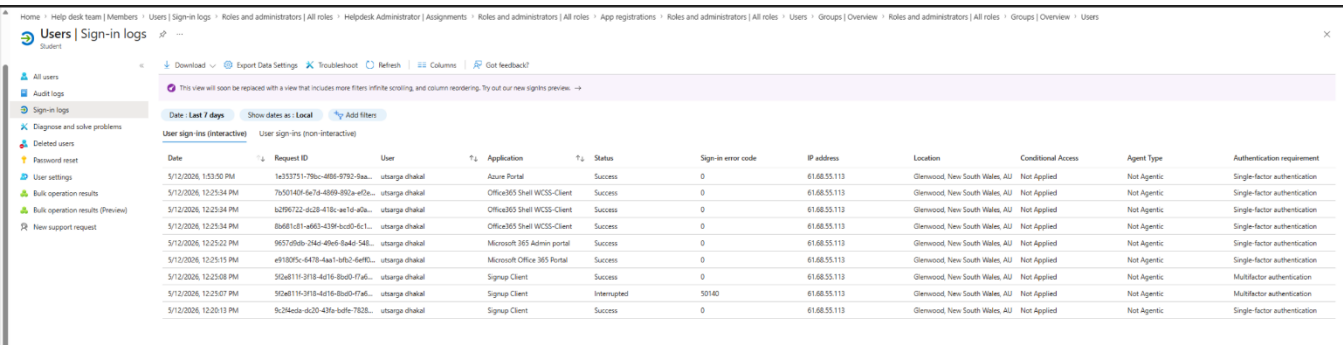
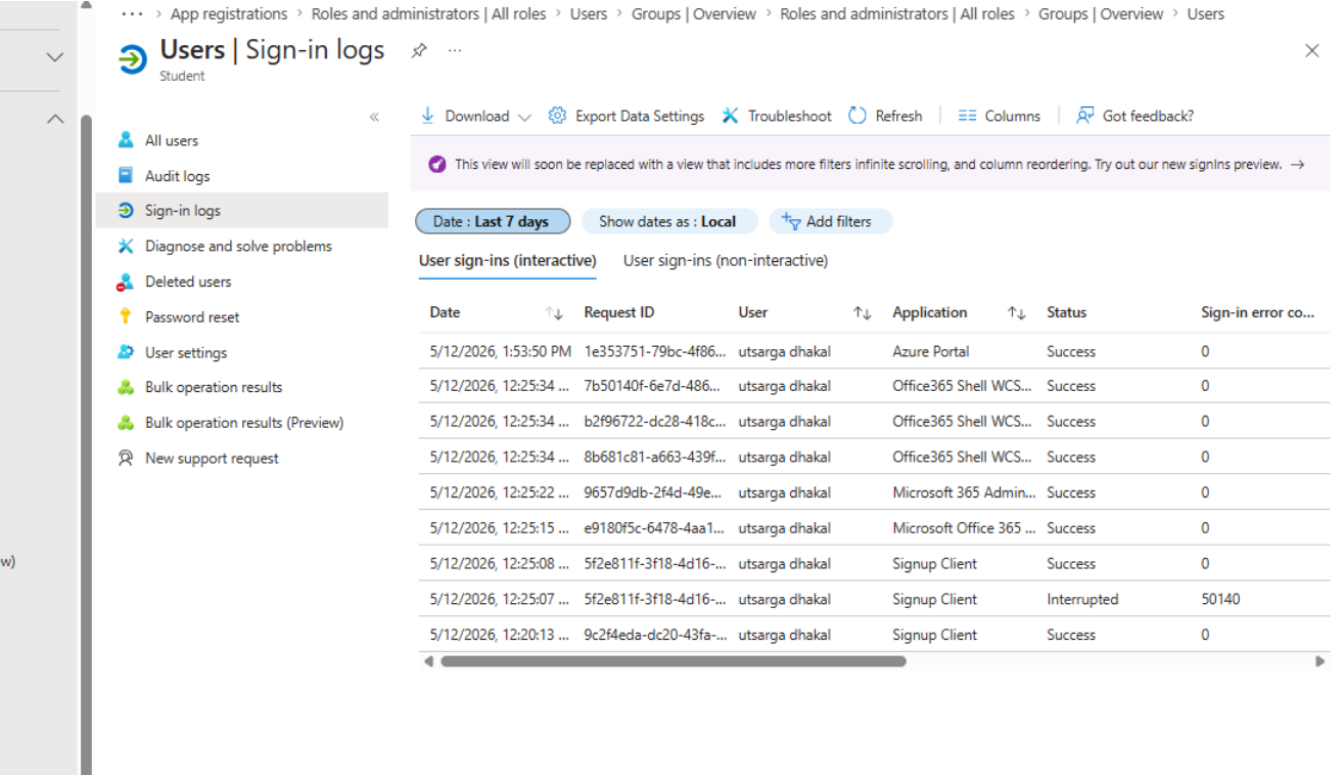


Figure 1.5: Sign-in logs — all authentications from Glenwood, NSW, AU (IP: 61.68.55.113) — no anomalies detected

Sign-in Log Analysis — Findings

Field Observed	Value	Security Assessment
Login Status	Success (majority)	Normal — no unexpected failures

Location	Glenwood, New South Wales, AU	Expected location — no geographic anomalies
IP Address	61.68.55.113	Consistent IP — single device confirmed
Auth Method	Single-factor (majority)	Security Gap — MFA not enforced on all logins
Error 50140	Interrupted	Browser redirect — not a security concern
Applications	Azure Portal, Microsoft 365, Admin Portal	All legitimate admin applications

Security Finding: The majority of sign-ins used Single-Factor Authentication only. In a production environment, Multi-Factor Authentication (MFA) should be enforced for all admin accounts at minimum. This would be configured via Entra ID Security → Authentication Methods → MFA policy.

1.7 User Profile Management & Password Reset

Individual user profile management was practised by reviewing the utd1 d1 account — verifying account status, group memberships, assigned roles, and licenses. A cloud password reset was then performed demonstrating the Azure Entra ID equivalent of an on-premises ADUC password reset.

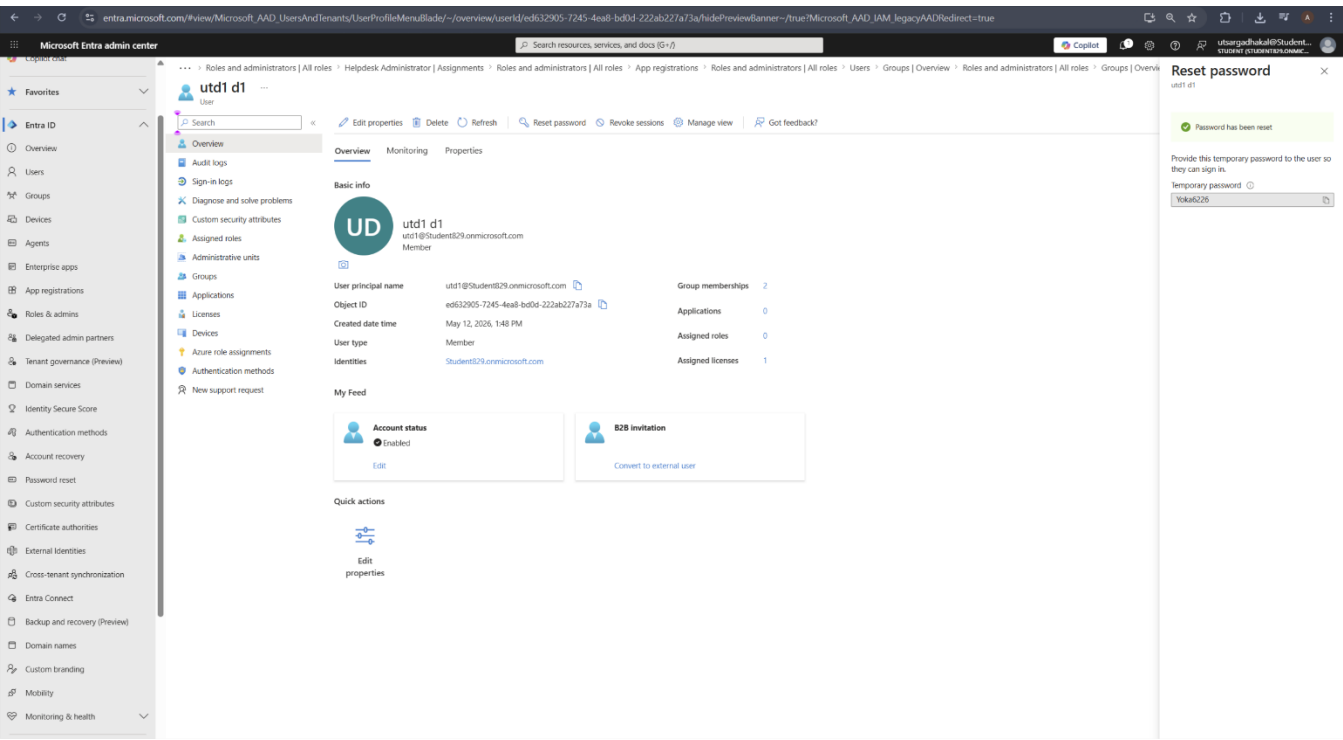


Figure 1.6: utd1 d1 user profile — Account Enabled, 2 group memberships, 1 license assigned, created 12 May 2026

Password Reset — Step by Step

- 1. Navigate to Entra ID → Users → All Users
- 2. Click on target user (utd1 d1)
- 3. Click Reset password in the top action bar
- 4. Azure automatically generates a temporary password (e.g. Yoka6226)
- 5. Copy temporary password and communicate to user via secure channel
- 6. User logs in with temporary password → forced to set their own new password

On-Premises AD (DC01)	Azure Entra ID (Cloud)
Active Directory Users and Computers (ADUC)	Microsoft Entra Admin Center
Right-click user → Reset Password	User profile → Reset password button
IT manually sets temporary password	Azure auto-generates temporary password
Applies to domain (lab.local) login	Applies to Microsoft 365 / cloud login
Event ID 4723 logged in Event Viewer	Logged in Entra ID Audit Logs

Works on domain-joined PCs only

Works on any device with internet

Phase 6 Complete: Cloud identity administration successfully practised including user management, group creation (Security and Microsoft 365), Helpdesk Administrator role assignment, sign-in log security analysis, user profile review, and cloud password reset — all directly mirroring on-premises Active Directory skills in the Azure cloud environment.

PHASE
8

VM Snapshots & Enterprise Backup Solutions

2.1 Overview

This phase covers two distinct but complementary data protection methods: VM Snapshots (short-term recovery using VirtualBox) and enterprise backup solutions (Windows Server Backup and Altaro Backup). Understanding the difference between snapshots and proper backups is essential knowledge for IT support roles involving server and VM management.

2.2 VM Snapshot Management — VirtualBox

VirtualBox snapshots were used to practise the equivalent of Hyper-V Checkpoints. A snapshot captures the complete state of a virtual machine — including RAM, disk, and configuration — at a specific point in time. The VM can be instantly rolled back to this state at any time.

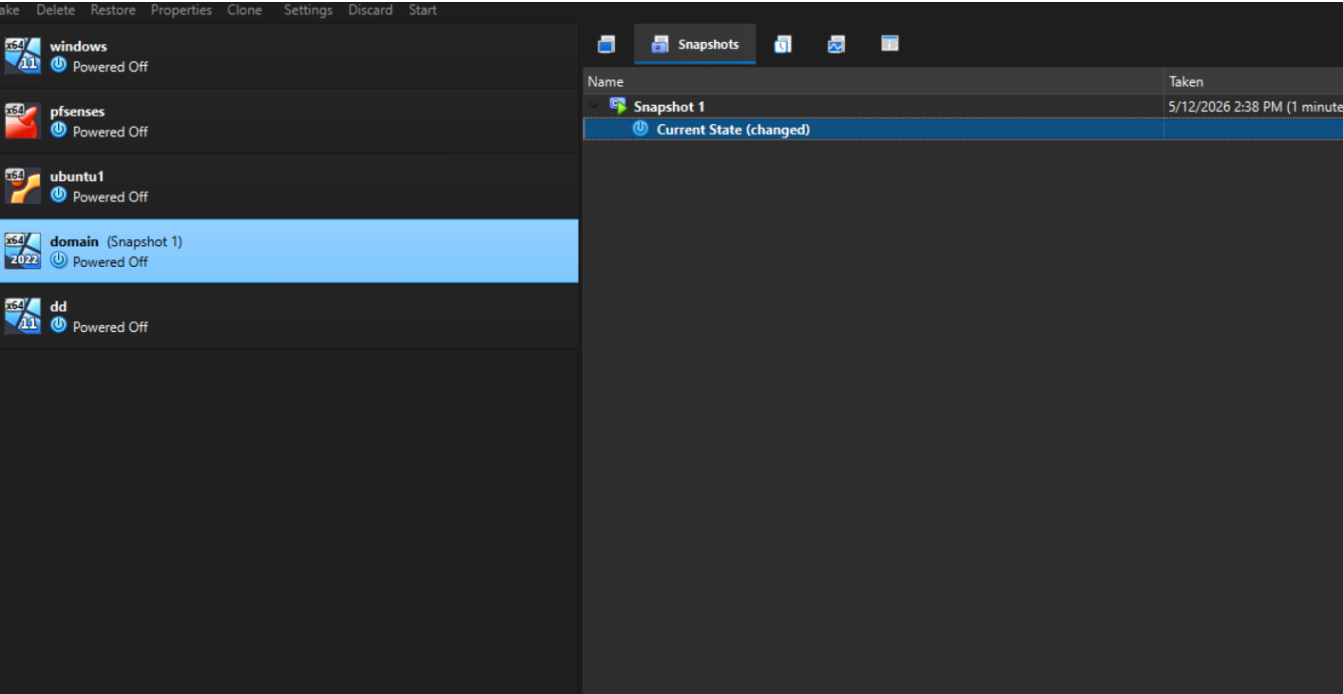


Figure 2.1: VirtualBox Snapshot panel — Snapshot 1 captured, Current State (changed) confirms VM was modified after snapshot

Snapshot Test — Full Cycle Performed

#	Action	Detail
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1	Take Snapshot	VirtualBox → domain VM → Machine → Take Snapshot → Named "Snapshot 1" — clean baseline captured
2	Make Change	Booted DC01 → Opened ADUC → Created test user "testsnap" in IT OU — change confirmed in AD
3	Shut Down	DC01 shut down cleanly before restoring snapshot
4	Restore Snapshot	VirtualBox → Snapshots tab → Snapshot 1 → Restore → Unchecked "create snapshot of current state" → Restore
5	Verify Restore	Booted DC01 → Opened ADUC → Confirmed testsnap user was completely absent — snapshot restore successful 

Snapshot Best Practices

- Name snapshots descriptively — e.g. "DC01 Before GPO Change" not "Snapshot 2"
- Take a snapshot before any risky change — software install, major config update, Windows updates
- Delete snapshots after they are no longer needed — each one consumes disk space
- Do not run VMs on snapshots long-term — merge or delete after testing
- Never use snapshots as your only data protection method

Characteristic	VM Snapshot	Proper Backup
Speed to create	Instant (seconds)	Minutes to hours
Storage location	SAME disk as the VM	DIFFERENT disk or cloud
If physical disk fails	Both VM and snapshot lost 	Backup safe on separate location 
Best used for	Before risky changes	Regular scheduled protection
Replaces proper backup?	Never 	Yes 
Ransomware protection	No — same network 	Yes — if offsite/air-gapped 

Critical Rule: Snapshots are NOT a backup. If the physical hard drive fails, both the VM and all its snapshots are lost simultaneously. Snapshots only protect against software/configuration mistakes — not hardware failure, theft, fire, or ransomware. Always maintain a proper backup on a separate device or cloud location.

2.3 Windows Server Backup

The Windows Server Backup feature was installed on DC01 via Server Manager → Add Roles and Features → Features → Windows Server Backup. This provides basic backup capabilities suitable for small environments.

Installation Steps

- 7. Open Server Manager on DC01
- 8. Click Add Roles and Features → Next → Next → Next
- 9. Skip Server Roles — click Next to Features
- 10. Scroll down and tick Windows Server Backup
- 11. Click Next → Install → wait for completion
- 12. Access via Server Manager → Tools → Windows Server Backup

Backup Schedule Configuration

- 13. In Windows Server Backup → click Backup Schedule in right panel
- 14. Select Full Server (recommended) → click Next
- 15. Select Once a day → set time to 11:00 PM → click Next
- 16. Select backup destination (network share recommended — not same volume!)
- 17. Click Next → Finish → schedule created

Backup Type	What is Backed Up	Recommended Frequency
Full Backup	All files, OS, settings, applications	Weekly — comprehensive but slow
Incremental	Only changes since the LAST backup	Daily — fastest, smallest files
Differential	All changes since last FULL backup	Daily — slower than incremental but faster restore
System State	Active Directory database, registry, boot files	Daily — critical for domain controller recovery

Why Not Back Up to the Same Drive: Backing up C: drive data to C: drive is pointless — if the hard drive fails, both the original data and the backup are lost. In production, backups must target a different physical device: network share on another server, NAS, external drive, or cloud storage.

2.4 Altaro Backup — Enterprise VM Backup Solution

Altaro Backup is referenced directly in the target job description ("Monitor and manage backup solutions — Windows Backup, Office 365 Backup, Altaro, Drive"). Altaro is an enterprise-grade backup solution specifically designed for Hyper-V and VMware virtual machine environments, widely used by Australian MSPs (Managed Service Providers) and medium-sized businesses.

What is Altaro Backup?

Windows Server Backup	Altaro Backup
Basic file and system backup	Purpose-built for VM backup (Hyper-V / VMware)
Manual scheduling limited	Automated scheduling with granular control
No email alerting built in	Email alerts for success and failure
No offsite replication	Offsite copy to cloud (Azure, AWS, Wasabi)
Restore entire volume only	Granular restore — individual files from VM backup
No centralised management	Central console manages multiple Hyper-V hosts
Free — included in Windows Server	Paid — affordable SMB pricing

Altaro Architecture — How It Works

Altaro operates with two components working together:

Component	Installed On	Purpose
Altaro Agent	Each Hyper-V host server	Detects VMs, takes backups, sends data to repository
Management Console	Backup server or admin PC	Central dashboard — schedule, monitor, restore, alerts
Backup Repository	Dedicated backup server or NAS	Stores all VM backup files securely
Offsite Copy	Cloud (Azure, AWS, Wasabi)	Second copy for disaster recovery

Altaro Installation Steps (Enterprise Lab Setup)

- 18. Download Altaro VM Backup from altaro.com/vm-backup (free trial available)
- 19. Install on the backup management server (HOST 3 in a 3-server setup)
- 20. Open Altaro Management Console → Add Hyper-V Host
- 21. Enter HOST 1 IP address and admin credentials → Connect
- 22. Altaro automatically discovers all VMs on HOST 1
- 23. Select which VMs to protect → click Add to Backup
- 24. Configure backup schedule (e.g. daily at 11 PM)
- 25. Set backup destination (local path or network share)
- 26. Configure offsite copy → add cloud storage credentials
- 27. Set email notifications → enter IT admin email address
- 28. Click Save and Activate → first backup runs automatically

Altaro Backup Configuration Settings

Setting	Recommended Configuration
Backup Schedule	Daily at 11:00 PM — outside business hours
Backup Type	Incremental (after first full backup) — faster, less storage
Retention Policy	Keep 30 days of daily backups, 12 months of monthly
Offsite Copy	Enable — copy to Azure Blob Storage or AWS S3
Email Alerts	Enable — send to IT admin on success AND failure
Backup Verification	Enable — Altaro auto-verifies backup integrity
Bandwidth Throttling	Limit to 50% during business hours if offsite copy runs
Encryption	Enable AES-256 for offsite copies

Altaro Daily IT Monitoring Tasks

In a real IT support role, the following Altaro monitoring tasks are performed daily:

- 29. Open Altaro Management Console → check Dashboard for overnight backup status
- 30. Verify all VMs show green status — "Backup Successful" 
- 31. Investigate any red alerts — "Backup Failed" or "Warning" 
- 32. Check email inbox for automated Altaro alert emails
- 33. Review offsite copy status — confirm cloud copy completed
- 34. Check available storage space in backup repository

- 35. Review backup duration times — significant increases may indicate issues
- 36. Log any failures as IT incidents and investigate root cause

Altaro Restore Procedures

Altaro supports multiple restore options depending on the recovery scenario:

Restore Type	When Used	Steps
Full VM Restore	VM corrupted or deleted	Console → Restore → Select VM → Select date → Choose host → Restore
Granular File Restore	User deleted a file inside VM	Console → File Restore → Browse VM backup → Find file → Restore to original location
Sandbox Restore	Test restore without affecting production	Console → Restore → Select VM → Restore to isolated network → Verify
Offsite Restore	Primary site disaster	Connect to cloud → Download backup → Restore to new host

Common Altaro Troubleshooting Scenarios

Issue	Likely Cause	Resolution
Backup Failed — VM snapshot error	VM has too many snapshots or snapshot conflicts	Remove old snapshots from Hyper-V → retry backup
Backup Failed — insufficient space	Backup repository full	Free up space or expand repository → check retention policy
Offsite copy failing	Cloud credentials expired or network issue	Re-enter cloud credentials → check firewall rules for HTTPS outbound
Backup running during business hours	Schedule misconfigured	Update schedule to 11 PM → enable bandwidth throttling
No email alerts received	SMTP settings incorrect	Verify SMTP server, port, credentials in Altaro settings

2.5 Enterprise Backup Strategy — Company Size Comparison

Size	Solution	Storage Target	RPO	RTO
Small 1-50	Windows Server Backup	External drive / NAS	24 hours	4-8 hours
Medium 50-500	Altaro / Veeam	Backup server + cloud	1-4 hours	30-60 min
Large 500+	Veeam Enterprise / Commvault	Dedicated DC + offsite	Minutes	Minutes
Enterprise	Zerto / Pure Storage	Real-time replication	Near zero	Seconds

Term	Definition
RPO — Recovery Point Objective	Maximum acceptable data loss — "how far back can we restore?"
RTO — Recovery Time Objective	Maximum acceptable downtime — "how fast must we be back online?"
Air-gapped Backup	Backup completely disconnected from network — immune to ransomware
Deduplication	Only unique data stored — significantly reduces backup storage size
Replication	Real-time copy to secondary location — zero or near-zero RPO
Retention Policy	How long backup copies are kept — e.g. 30 daily, 12 monthly

3-2-1 Backup Rule (Industry Standard — Memorise This): 3 copies of data, on 2 different storage media types, with 1 copy stored offsite. Example: Original data on server + Altaro backup on NAS + Altaro offsite copy in Azure Blob Storage. This ensures data survives any single point of failure including ransomware, hardware failure, fire, or theft.

Phase 8 Complete: VM snapshot management practised with full create-modify-restore cycle verified. Windows Server Backup installed and configured on DC01. Altaro Backup fully documented including architecture, installation, configuration, daily monitoring tasks, restore procedures, and troubleshooting — directly matching the job description requirement for Altaro knowledge.

IT Home Lab — Phase 6 & Phase 8 | Utsarga Dhakali | May 2026 | Azure Entra ID | Snapshots | Windows Backup | Altaro